DOMAIN-RELEVANT SOFTWARE EVOLUTION

A Case Study of Covasim

Yamir Poldo Silva
Université de Montréal
poldosiy@iro.umontreal.ca

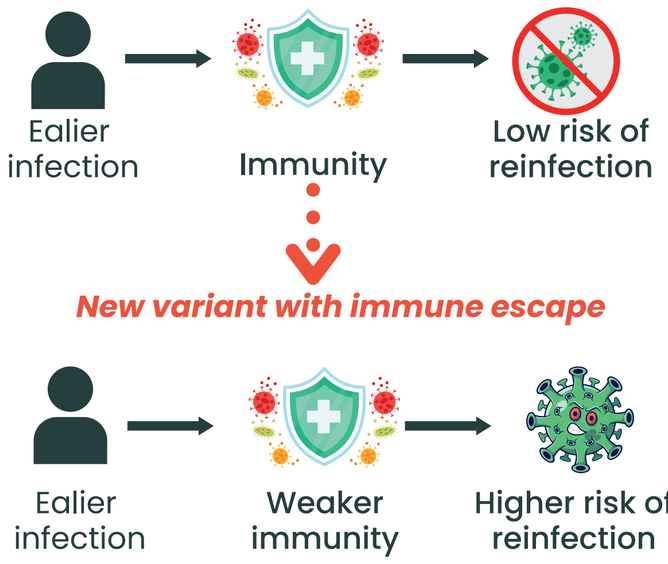
Michalis Famelis
Université de Montréal
famelis@iro.umontreal.ca

How does domain-relevant development evolve following a domain change, and how unusual is that evolution relative to the project's historical development?

1 MOTIVATION

Domain knowledge evolves

New evidence can change what software is expected to represent



Software tests are not enough

Software may still run and pass tests while its encoded domain assumptions become outdated

Current domain Knowledge
Updated scienctif understanding (e.g. immune escape, variant properties, cross-immunity)

domain-assumption drift

Assumptions represented in software
Encoded model structure, parameters and relationship (e.g. immunity, susceptibility)

Can repository reveal it?

Repositttory activity may provide clues that represented assumptions are changing.

- **Jan. 19, 2021** — **New evidence of immune escape**
  Scientific event (external anchor)
- **Feb. - Mar.** — **Repeated immunity-related changes**
  Multiple commits related to rewoke of immunity, susceptibility, etc.
- **Mar. 3** — **Immunity becomes an explicit software module**
  immunity.py

The scientific event is an external anchor. This sequence is not interpreted as evidence of causality.

2 PRELIMINARY APPROACH

CASE STUDY - COVASIM | Agent-based COVID-19 modeling software in the broader Starsim infectious-disease modeling ecosystem.

01 EXTERNAL EVENT
Jan. 19, 2021
New evidence on immune escape in COVID-19 variants

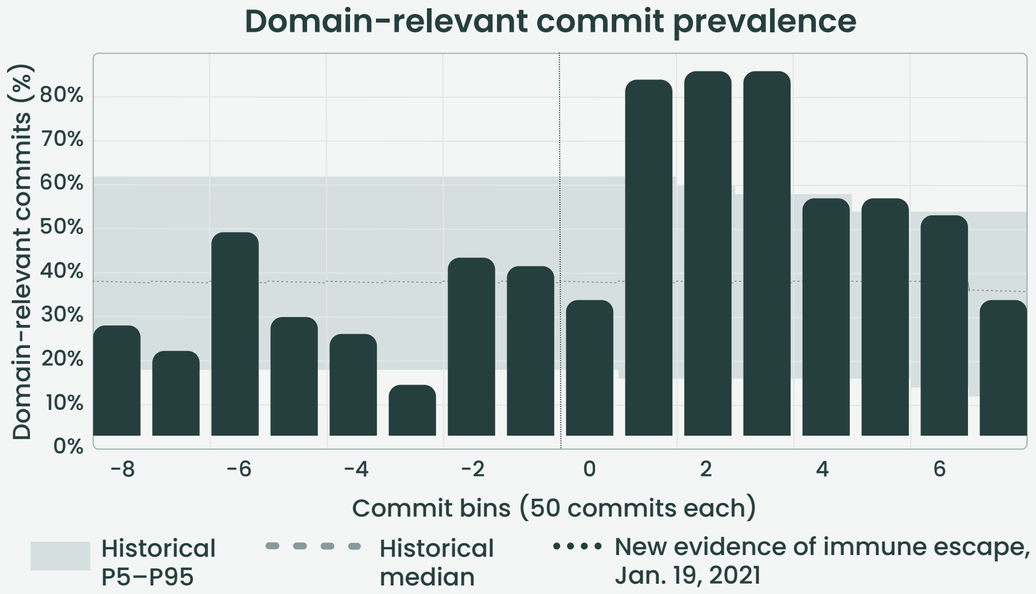
02 DOMAIN CLASSIFICATION
Domain-relevant vs. non-domain relavant commits
Project-specific criteria + LLM classifier.

03 REPOSITORY HISTORY
Analyze 400 commits before and after the event

04 HISTORICAL CALIBRATION
Compare with historical pseudo-events from the same repository.

3 PRELIMINARY FINDINGS | Domain-Relevant Development Became Unusually Concentrated

Domain-relevant commit prevalence



85.3% **of commits in Bins 1–3 were domain-relevant**
128 / 150 commits

Historically unusual trajectory. Bins 1-3 exceeced the historical 95th percentile; elevated domain prevalence above 90th percentile throught Bin 5.

What was happening during this unusual episode?

Preliminary inspection of immunity-related representative commits

Bin 1 · Feb. 5–16, 2021 - Representation expands

Feb. 8 - 9cdb8ac - making immunity strain-specific
Feb. 10 - 158dac0 - immunity … is now a matrix
Feb. 16 - 1dac390 - "stuck on dimensions"

Intertepration: Strain-specific immunity and cross-immunity become explicit, with early implementation friction.

Bin 2 · Feb. 16-Mar 1. - Representation is reworked

Feb. 18 - b46787c - "scenarios need refactoring "
Feb. 22 - 26dd3b5 - "ugly fix for update_immunity…"
Feb. 25 - cc403df - "trying to simplify strain information…"

Intertepration: Immunity expands into multiple effects while strain and scenario logic is reworked.

Bin 3 · Mar. 1-Mar 10. - Representation is consolidated

Mar. 1 - 336420b - "merged immunity and cross immunity calculations…"
Mar 2. 7d377c8 - "stuck with immunity calcs"
Mar. 3 - 46c4070 - "add immunity file"

Intertepration: Immunity calculations are consplidated into an explicit module.

Across Bins 1–3, immunity representation is repeatedly extended, reworked and eventually consolidated into an explicit module.
Case-study caveat: One retrospective Covasim episode; no causal or general claim.

4 NEXT STEPS

RQ1 Detecting Domain Assumption Drift

Can repository artifacts reveal domain-assumption drift early enough to support software evolution planning?

Recover domain representations across versions and identify repository signals of changing assumptions.

RQ2 Characterizing and Communicating Domain Assumption Drift

What information helps stakeholders understand a candidate drift signal and assess its implications for software evolution?

Study which representation best support stakeholder understanding of changed assumptions

Help shape the next study

We are recruiting participants who use programming or computational tools in their work.

Interested in participating or sharing feedback?

[QR CODE]